Md. Jubair Uddin Zidan

Mechanical Engineering Researcher • Computational Fluid Dynamics & Architected Metamaterials

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EDUCATION

Master of Science (M.Sc.) in Mechanical Engineering
Chittagong University of Engineering and Technology (CUET)
2025 – 2027
Ongoing Postgraduate

- Theory Coursework Cumulative CGPA: **3.90 / 4.00**
- Master's Thesis Research in Progress (Tentative Defence: January 2027)

Bachelor of Science (B.Sc.) in Mechanical Engineering
Chittagong University of Engineering and Technology (CUET)
2019 – 2024
Degree Conferred

- Cumulative GPA across Upper Semesters: **3.41 / 4.00** (Graduating Class: 2024)
- Senior Capstone Thesis: High-Fidelity 3D Ground Effect Aerodynamics of Modern F1 Venturi Tunnels

RESEARCH EXPERIENCE & PUBLICATIONS

Undergraduate Bachelor Thesis: Aerodynamics & Ground Effect CFD
Chittagong University of Engineering and Technology (CUET), Dept. of Mechanical Engineering | 2024

- Conducted full-domain 3D computational fluid dynamics (CFD) simulations investigating Venturi ground-effect floors under modern FIA Formula One technical regulations.
- Analyzed vortex breakdown phenomena, suction peak preservation, downforce generation, and sensitivity to ride-height variations.

Current Manuscript: Architected Auxetic Mechanical Metamaterials for Energy Absorption
Under Review | Authors: Jubair Zidan, Wakil Ahmed, Dibosh Das | Supervisor: Dr. Kazi Zahir Uddin (UIUC / CUET)

- Systematic state-of-the-art review and numerical characterization of negative Poisson's ratio cellular metamaterials (2D re-entrant, chiral, and 3D rotational lattices).
- Evaluated specific energy absorption (SEA), dynamic impact crashworthiness, and multi-objective topology optimization.

Conference Paper: Aerodynamic Characterization of Bionic Tubercles on Airfoils
International Conference on Mechanical Engineering and Renewable Energy (ICMERE)

- Investigated stall delay mechanisms and boundary layer vortex generation across modified NACA airfoils under varying Reynolds numbers.

TECHNICAL & COMPUTATIONAL SKILLS

CAD & Design:	SolidWorks
Simulation & CFD:	ANSYS Fluent, ANSYS LS-DYNA, OpenFOAM (Foundational Knowledge), COMSOL Multiphysics (Foundational Knowledge)
Programming:	MATLAB, Python, C (Coursework), Fortran (Coursework)
Data Analysis & Computing:	NumPy, Pandas, Jupyter Notebook, OriginPro, MS Excel
Documentation & Graphics:	Git & GitHub, LaTeX, Mendeley, MS Office, Google Suite, Adobe Suite, Figma
Operating Systems:	macOS, Windows

PROFESSIONAL & LEADERSHIP EXPERIENCE

ASME Student Section CUET — Executive Committee Member (2022 – 2024)

- Organized technical symposia, CAD modeling challenges, and national student engineering conventions.

Industrial Engineering Internship — Thermal Power Generation Plant (2023)

- Hands-on attachment analyzing combined cycle gas turbine operations, heat recovery steam generators (HRSG), and maintenance protocols.

Academic Instructorship & STEM Mentor — Physics & Higher Mathematics (2021 – Present)

- Taught Newtonian mechanics, electromagnetism, and advanced calculus to aspiring undergraduate engineering candidates.

HONORS, AWARDS & RECOGNITION

- **Champion — National Mechanical Engineering Innovation Design Challenge**

- **Scholarly Poster Symposium Award** for CFD Ground-Effect Investigations
- **Global Student Leadership Summit Delegate** representing CUET Faculty of Mechanical Engineering